

CT214H – Web Programming

Practical Session 3 – JavaScript

Content

- ❖ Using JavaScript to program the web on the browser side:
 - Using JS to process events and perform calculations.
 - Combine JS, DOM, and CSS to style the webpage dynamically.

Ex 1. Do the examples in the lecture.

- ❖ **Objective:**
 - Learn the JavaScript programming language.
 - Use JavaScript and the DOM to modify the page content and format
- ❖ **Requirement:**
 - Do the examples in the lecture, browse them, and observe the results.
 - Modify the examples and observe the results.

Ex2. Builds interactive features on the TechBlog webpage

- ❖ **Objective:** Building interactive features on the TechBlog webpage. Your starting point should be from the exercise 3 of the previous practical session.
- ❖ **Preparation steps:**
 1. Create a new JavaScript file “techblog.js”
 2. Link JavaScript to HTML: before the end of body
`<script src="techblog.js"></script>`
- ❖ **Requirement:** step-by-step adding the features as describe in the tasks below and observe the results

1. Search filter

Create a search box that filters articles in real-time as the user types: Search titles and excerpts; Case-insensitive search.

a. Add Search Box to HTML before the main content area

```
<div style="max-width: 1200px; margin: 30px auto; padding: 0 20px;">
  <input type="text" id="searchInput" placeholder="🔍 Search articles..."
    style="width: 100%; padding: 15px; font-size: 1.1rem; border-radius: 8px; border: 2px solid #ddd; box-shadow: 0 2px 5px rgba(0,0,0,0.1);">
</div>
```

b. Add event listener in your JavaScript file

Get search input element

```
const searchInput = document.getElementById('searchInput');
```

Add event listener for typing

```
searchInput.addEventListener('input', initSearchFilter);
```

c. The *initSearchFilter* function

Get all article cards

```
const articleCards = document.querySelectorAll('.article-card');
```

Get search text in lowercase

```
const searchTerm = searchInput.value.toLowerCase();
```

Loop through each article card

```
articleCards.forEach(function(card) {  
  // Get title and excerpt text  
  const title = card.querySelector('.article-title').textContent.toLowerCase();  
  const excerpt = card.querySelector('.article-excerpt').textContent.toLowerCase();  
  // Check if search term is found  
  if (title.includes(searchTerm) || excerpt.includes(searchTerm)) {  
    card.style.display = ""; // Show card  
  } else {  
    card.style.display = 'none'; // Hide card  
  }  
});
```

d. Observe the results

- Type in search box to filter articles; Search words on titles and excerpts

2. Dark Mode Toggle

Switch light/dark themes

a. Add this HTML inside your navbar <ul class="nav-menu">

```
<li>  
  <button id="darkModeToggle" style="background: none; border: none; color: white; font-size: 1.5rem;  
  cursor: pointer;"> 🌙 </button>  
</li>
```

b. Add these CSS styles to your CSS file

```
body.dark-mode {  
  background-color: #1a1a1a;  
  color: #e0e0e0;  
}  
body.dark-mode .article-card, body.dark-mode .featured-article,  
body.dark-mode .sidebar, body.dark-mode .widget {  
  background-color: #2c2c2c;  
  color: #e0e0e0;  
}  
body.dark-mode h2, body.dark-mode h3 {  
  color: #ffffff;  
}  
body.dark-mode .popular-list a, body.dark-mode .category-list a {  
  color: #e0e0e0;  
}
```

c. Add event listener in your JavaScript file

Get the darkModeToggle button

```
const darkModeToggle = document.getElementById('darkModeToggle');
```

Toggle dark mode on button click

```
darkModeToggle.addEventListener('click', initDarkMode);
```

d. The initDarkMode function

```
document.body.classList.toggle('dark-mode');
// Update icon and save preference
if (document.body.classList.contains('dark-mode')) {
  darkModeToggle.textContent = '☀️';
  localStorage.setItem('darkMode', 'enabled');
} else {
  darkModeToggle.textContent = '🌙';
  localStorage.removeItem('darkMode');
}
```

e. Check user's preference

You need localStorage so the browser can remember the user's choice after the page is reloaded or the browser is closed: localStorage stores data inside the browser.

```
// Check if dark mode was previously enabled
if (localStorage.getItem('darkMode') === 'enabled') {
  document.body.classList.add('dark-mode');
  darkModeToggle.textContent = '☀️';
}
```

f. Observe the results

- Click on the darkMode button and observe the result
- Reload page after change to dark mode

3. Smooth Scroll to Section

Clicking nav links smoothly scrolls to sections

a. Modify your HTML

```
<!-- Make sure your nav links have href to section IDs -->
<nav class="navbar">
  <ul class="nav-menu">
    <li><a href="#home">Home</a></li>
    <li><a href="#about">About</a></li>
    <li><a href="#articles">Articles</a></li>
  </ul>
</nav>
<!-- Add IDs to your sections -->
<header class="hero" id="home">
<section id="articles" class="articles">...</section>
<footer class="footer" id="about">
```

b. Add event listener to your JavaScript file

Select the navigation links in navigation bar

```
const navLinks = document.querySelectorAll('.nav-menu a');
```

Add event listener when user click on the links

```
navLinks.forEach(function(link) {
  link.addEventListener('click', initSmoothScroll)
});
```

c. The `initSmoothScroll` function

- `preventDefault`: Prevents the browser from instantly jumping to the URL in the `href` attribute
- Get the target section
- Smooth scroll to the section

```
function initSmoothScroll(e) {  
  e.preventDefault(); //Prevent default anchor behavior  
  const targetId = this.getAttribute('href');  
  const targetSection = document.querySelector(targetId);  
  
  if (targetSection) {  
    targetSection.scrollIntoView({  
      behavior: 'smooth',  
      block: 'start'  
    });  
  }  
}
```

d. Observe the results

- Click links on the navigation bar and observe the results

4. Back to Top Button

Appears when scrolling down

a. Add this HTML at the end of your body (before `</body>`)

```
<button id="backToTop" style="position: fixed; bottom: 30px; right: 30px; background-color:  
#3498db; color: white; border: none; border-radius: 50%; width: 50px; height: 50px; font-size:  
1.5rem; cursor: pointer; display: none; box-shadow: 0 4px 10px rgba(0,0,0,0.3);">  
  ↑ Hello  
</button>
```

b. Add event listener in your JavaScript file

Get the `backToTop` button

```
const backToTopBtn = document.getElementById('backToTop');
```

Scroll to top when clicked

```
backToTopBtn.addEventListener('click', initBackToTop);
```

c. The `initBackToTop` function

```
window.scrollTo({  
  top: 0,  
  behavior: 'smooth' // Smooth scroll animation  
});
```

d. Show/hide button based on scroll position

```
window.addEventListener('scroll', function() {  
  if (window.pageYOffset > 300) {  
    backToTopBtn.style.display = 'block';  
  } else {  
    backToTopBtn.style.display = 'none';  
  }  
});
```

e. Observe the results

- Scroll the page and click on the button to back to the top

5. Article View Counter

Track how many times each article is viewed (clicked)

a. Add this HTML inside of each article-body, before the closing div

```
<p class="view-count" style="color: #999; font-size: 0.9rem; margin-top: 10px;">  <span  
class="views">0</span> views  
</p>
```

b. Add event listener in your JavaScript file

Get all article-card elements

```
const articleCards = document.querySelectorAll('.article-card');
```

Loop through each article-card

```
articleCards.forEach(function(card, index) {  
  const viewSpan = card.querySelector('.views');  
  // Load saved view count  
  const views = localStorage.getItem(`article-${index}-views`) || 0;  
  viewSpan.textContent = views;  
  // Increment view count when article is clicked  
  card.addEventListener('click', function() {  
    const currentViews = parseInt(viewSpan.textContent);  
    const newViews = currentViews + 1;  
    viewSpan.textContent = newViews;  
    localStorage.setItem(`article-${index}-views`, newViews);  
  });  
});
```

6. Sort articles

Sort articles by newest, oldest

a. Add this HTML inside of the articles section, before articles grid

```
<div class="sort-controls" style="margin: 20px 0;">
  <label>Sort by:</label>
  <select id="sortSelect" style="padding: 8px; border-radius: 5px;">
    <option value="default">Default</option>
    <option value="newest">Newest First</option>
    <option value="oldest">Oldest First</option>
    <option value="popular">Most View</option>
  </select>
</div>
```

b. Add event listener in your JavaScript file

Get the dropdown box for order selection

```
const sortSelect = document.getElementById('sortSelect');
```

Add event listener on selection changing

```
sortSelect.addEventListener('change', initSort);
```

c. Save the grid appearing original order (HTML order)

```
const articlesGrid = document.querySelector('.articles-grid');
const originalArticles = Array.from(
  articlesGrid.querySelectorAll('.article-card'));
```

d. The initSort function

```
const sortBy = this.value; //this refers to the <select> element
let articles = [];
if (sortBy === 'default') {
  // Restore original HTML order
  articles = [...originalArticles]; //Spread Operator create a copy
} else {
  articles = Array.from(document.querySelectorAll('.article-card'));
  if (sortBy === 'newest') {
    articles.sort((a, b) => {
      const dateA = new Date(a.querySelector('.date')?.textContent.trim());
      const dateB = new Date(b.querySelector('.date')?.textContent.trim());
      return dateB - dateA; }); } //if
  if (sortBy === 'oldest') {
    articles.sort((a, b) => {
      const dateA = new Date(a.querySelector('.date')?.textContent.trim());
      const dateB = new Date(b.querySelector('.date')?.textContent.trim());
      return dateA - dateB; }); } //if
  if (sortBy === 'popular') {
    articles.sort((a, b) => {
      const viewA = parseInt(a.querySelector('.views')?.textContent);
      const viewB = parseInt(b.querySelector('.views')?.textContent);
      return viewB - viewA; }); } //if
  } // Re-append articles in new order
  articles.forEach(article => articlesGrid.appendChild(article));
```

d. Observe the results

- Changing the articles by sort selection and observe the results

7. Newsletter Popup

Show newsletter signup popup after 5 seconds

a. Add this HTML at the end of your body (before </body>)

```
<div id="newsletterPopup" style="display: none; position: fixed; bottom: 20px; right: 20px; width: 350px; background: white; padding: 25px; border-radius: 10px; box-shadow: 0 5px 20px rgba(0,0,0,0.3); z-index: 9999;">
  <span id="closePopup" style="position: absolute; top: 10px; right: 15px; font-size: 1.5rem; cursor: pointer;">&times;</span>
  <h3>📧 Subscribe to TechBlog</h3>
  <p>Get the latest tech news delivered weekly!</p>
  <input type="email" id="popupEmail" placeholder=" Your email address" style="width: 100%; padding: 10px; margin: 10px 0; border: 1px solid #ddd; border-radius: 5px;">
  <button id="subscribeBtn" style="width: 100%; padding: 10px; background: #3498db; color: white; border: none; border-radius: 5px; cursor: pointer;">Subscribe</button>
</div>
```

b. In your JavaScript file

Get the elements by ID

```
const popup = document.getElementById('newsletterPopup');
const closePopup = document.getElementById('closePopup');
const subscribeBtn = document.getElementById('subscribeBtn');
const emailInput = document.getElementById('popupEmail');
```

Check if user already subscribed or closed: display popup after 5 seconds

```
const hasSubscribed = localStorage.getItem('newsletter-subscribed');
const hasClosed = localStorage.getItem('newsletter-closed');
if (!hasSubscribed && !hasClosed) {
  setTimeout(function() {
    popup.style.display = 'block';
  }, 5000); // Show after 5 seconds
}
```

c. Add event listener when user close popup

```
closePopup.addEventListener('click', closeNewsletterPopup);
```

The closeNewsletterPopup function

```
popup.style.display = 'none';
localStorage.setItem('newsletter-closed', 'true');
```

d. Add event listener when user subscribe

```
subscribeBtn.addEventListener('click', initNewsletterPopup);
```

The initNewsletterPopup function

```
const email = emailInput.value;
if (email.includes('@')) {
  alert('Thank you for subscribing! 🎉');
  popup.style.display = 'none';
  localStorage.setItem('newsletter-subscribed', 'true');
} else {
  alert('Please enter a valid email address');
}
```

e. Observe the results

- Observe the popup after page load